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843. On the Quadriquadric Curve in connexion with the Theory of Elliptic Functions.

[From the *Mathematische Annalen*, t. XXV. (1885), pp. 152–156.]

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I CALL to mind that, if we have on a line two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and through the line two planes

$$Ax + By + Cz + Dw = 0, \quad A'x + B'y + C'z + D'w = 0,$$

then

$$\begin{aligned} \beta\gamma' - \beta'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta : \alpha\delta' - \alpha'\delta : \beta\delta' - \beta'\delta : \gamma\delta' - \gamma'\delta \\ = AD' - A'D : BD' - B'D : CD' - C'D : BC' - B'C : CA' - C'A : AB' - A'B; \end{aligned}$$

and that, putting each of these two equal sets of ratios

$$= a : b : c : f : g : h,$$

then the quantities (a, b, c, f, g, h) , which it is easy to see satisfy the relation

$$af + bg + ch = 0,$$

are said to be the six “coordinates” of the line: as only the ratios of the six quantities are material, and as the last-mentioned equation establishes a single relation between these ratios, the system of the six coordinates contains four arbitrary ratios or parameters, for the determination of the particular line. See my paper “On the six coordinates of a line,” *Camb. Phil. Trans.*, t. XI. (1869), pp. 290–323, [435].

I consider for a moment the quadric surface

$$x^2 + y^2 + z^2 + w^2 = 0,$$

and I proceed to show that, if (a, b, c, f, g, h) are the coordinates of a generating line on the surface, then either $a = f$, $b = g$, $c = h$, or else $a = -f$, $b = -g$, $c = -h$; the one or other system of equations according as the line belongs to the one or other system of generating lines.

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We satisfy the equation by

$$\begin{aligned} x + iy + \theta z + \theta iw &= 0, \\ x - iy - \frac{1}{\theta}z + \frac{1}{\theta}iw &= 0, \end{aligned}$$

where θ is an arbitrary parameter: hence these equations determine a generating line of the surface: and the coordinates of this line are

$$= -i \begin{pmatrix} a, \\ \left(\theta - \frac{1}{\theta}\right) \end{pmatrix}, \quad - \begin{pmatrix} b, \\ \left(\theta + \frac{1}{\theta}\right) \end{pmatrix}, \quad 2i, \quad i \begin{pmatrix} f, \\ \left(\theta - \frac{1}{\theta}\right) \end{pmatrix}, \quad \theta + \frac{1}{\theta}, \quad -2i;$$

viz. these values give $a = -f$, $b = -g$, $c = -h$.

Similarly we satisfy the equation by

$$x + iy + \phi z - \phi iw = 0,$$

$$x - iy - \frac{1}{\phi}z - \frac{1}{\phi}iw = 0,$$

where ϕ is an arbitrary parameter: hence these equations also determine a generating line of the surface: and the coordinates of this line are

$$= i \begin{pmatrix} a, & b, & c, & f, & g, & h \\ \left(\phi - \frac{1}{\phi}\right), & \phi + \frac{1}{\phi}, & -2i, & i\left(\phi - \frac{1}{\phi}\right), & \phi + \frac{1}{\phi}, & -2i; \end{pmatrix}$$

viz. these values give $a = f$, $b = g$, $c = h$.

If for x, y, z, w we write $x\sqrt{p}$, $y\sqrt{q}$, $z\sqrt{r}$, $w\sqrt{s}$ respectively, then we have the theorem that, for the quadric surface

$$px^2 + qy^2 + rz^2 + sw^2 = 0,$$

the coordinates (a, b, c, f, g, h) of a generating line are such that

$$\frac{a}{f} = \pm \sqrt{\frac{ps}{qr}}, \quad \frac{b}{g} = \pm \sqrt{\frac{qs}{rp}}, \quad \frac{c}{h} = \pm \sqrt{\frac{rs}{pq}},$$

the signs being all + or all −, according as the line belongs to one or other of the systems of generating lines.

Take (a', b', c', f', g', h') for the coordinates of an arbitrary line, and write

$$p, q, r, s = a'g'h', b'h'f', c'f'g', a'b'c';$$

the quadric surface is

$$a'g'h'x^2 + b'h'f'y^2 + c'f'g'z^2 + a'b'c'w^2 = 0,$$

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which is a surface having the line (a', b', c', f', g', h') for a generating line. To verify this, observe that for the line in question we have

$$\begin{aligned} h'y - g'z + a'w &= 0, \\ -h'x + f'z + b'w &= 0, \\ g'x - f'y + c'w &= 0, \\ -a'x - b'y - c'z &= 0, \end{aligned}$$

equivalent of course to two independent equations: these give $h'x = f'z + b'w$, $h'y = g'z - a'w$, values which substituted in the quadric equation satisfy it identically. And for the last-mentioned quadric surface we have the theorem that, if (a, b, c, f, g, h) are the coordinates of a generating line, then

$$\frac{a}{f} = \pm \frac{a'}{f'}, \quad \frac{b}{g} = \pm \frac{b'}{g'}, \quad \frac{c}{h} = \pm \frac{c'}{h'},$$

where obviously the sign + belongs to a generating line of the same system with the line (a', b', c', f', g', h') , and the sign − to a line of the other system.

Taking the sign −, we thus see that if (a, b, c, f, g, h) , (a', b', c', f', g', h') are the coordinates of lines of the two systems respectively, then

$$af' + a'f = 0, \quad bg' + b'g = 0, \quad ch' + c'h = 0;$$

where observe that the resulting equation

$$af' + a'f + bg' + b'g + ch' + c'h = 0,$$

is the condition which expresses that the two lines meet each other.

Consider now the quadriquadric curve

$$U_1 = Ax^2 + By^2 + Cz^2 + Dw^2 = 0,$$

$$U_2 = A'x^2 + B'y^2 + C'z^2 + D'w^2 = 0;$$

and let (a, b, c, f, g, h) and (a', b', c', f', g', h') be the coordinates of two lines meeting each other, and each meeting the quadric curve twice: or, again, let these be lines joining in pairs the four intersections of the curve by an arbitrary plane: or, again, let them be the nodal lines of the binodal quartic cone having an arbitrary vertex and passing through the curve. The two lines are generating lines, belonging to the two systems respectively, of a properly determined quadric surface $U_1 + \lambda U_2 = 0$ passing through the curve: and by what precedes, we have

$$fa' + af' = 0, \quad bg' + b'g = 0, \quad ch' + c'h = 0,$$

the fundamental theorem which I wished to establish.

Writing in the equations $w = 1$, we have in particular the quadriquadric curve $y^2 = 1 - x^2$, $z^2 = 1 - k^2x^2$, equations which are satisfied by $x = \text{sn } u$, $y = \text{cn } u$, $z = \text{dn } u$. Consider on the curve four points belonging to the arguments u_1, u_2, u_3, u_4 respectively;

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and write for shortness s_1, c_1, d_1 for the sn, cn, and dn of u_1 ; and similarly for u_2, u_3 and u_4 . It appears by Abel's theorem that the condition in order that the four points may be in a plane is $u_1 + u_2 + u_3 + u_4 = 0$; viz. when this equation is satisfied we have

$$\begin{vmatrix} s_1 & c_1 & d_1 & 1 \\ s_2 & c_2 & d_2 & 1 \\ s_3 & c_3 & d_3 & 1 \\ s_4 & c_4 & d_4 & 1 \end{vmatrix} = 0,$$

or writing

$$\begin{aligned} & \begin{matrix} a, & b, & c, & f, & g, & h \\ = c_1d_2 - c_2d_1, & d_1s_2 - d_2s_1, & s_1c_2 - s_2c_1, & s_1 - s_2, & c_1 - c_2, & d_1 - d_2, \\ a', & b', & c', & f', & g', & h' \\ = c_3d_4 - c_4d_3, & d_3s_4 - d_4s_3, & s_3c_4 - s_4c_3, & s_3 - s_4, & c_3 - c_4, & d_3 - d_4, \end{matrix} \end{aligned}$$

this equation is

$$af' + a'f + bg' + b'g + ch' + c'h = 0;$$

or by what precedes, it appears that not only is this so, but that we have separately

$$af' + a'f = 0, \quad bg' + b'g = 0, \quad ch' + c'h = 0,$$

viz. it follows from Abel's theorem that, when the arguments u_1, u_2, u_3, u_4 are connected by the equation $u_1 + u_2 + u_3 + u_4 = 0$, then each of these three equations holds good.

I assume in particular $u_1 = 0$, $u_3 = -u$, $u_4 = u$, so that $u = u_1 + u_2$; we have

$$\begin{aligned} & \begin{matrix} a, & b, & c, & f, & g, & h \\ = c_1d_2 - c_2d_1, & d_1s_2 - d_2s_1, & s_1c_2 - s_2c_1, & s_1 - s_2, & c_1 - c_2, & d_1 - d_2, \\ a', & b', & c', & f', & g', & h' \\ = cd, & s, & -s, & -s, & c - 1, & d - 1, \end{matrix} \end{aligned}$$

and the three equations become

$$\frac{c_1d_2 - c_2d_1}{s} = \frac{d_1s_2 - d_2s_1}{c_1 - c_2} = \frac{s_1c_2 - s_2c_1}{d_1 - d_2},$$

or, what is the same thing,

$$\frac{c_1d_2 - c_2d_1}{s} = \frac{-c + 1}{s} = \frac{d - 1}{s},$$

equations which may also be written

$$\frac{c_1 d_2 - c_2 d_1}{s} = \frac{-c + 1}{s} = \frac{d - 1}{s} = \frac{c_1 - c_2}{d_1 s_2 - d_2 s_1} = \frac{d_1 - d_2}{s_1 c_2 - s_2 c_1},$$

so that, adding these three equations, we must have an identity.

Representing the second and third of the last-mentioned equations by

$$\frac{-c + 1}{s} = C, \quad \frac{d - 1}{s} = D,$$

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we have

$$c = 1 - Cs, \quad d = 1 + Ds,$$

from either of which we can obtain s rationally; viz. the first equation gives

$$1 - s^2 = 2Cs + C^2 s^2,$$

that is, $s = \frac{2C}{1 + C^2}$, whence

$$c = \frac{1 - C^2}{1 + C^2}, \quad d = \frac{1 + C^2 + 2CD}{1 + C^2};$$

and similarly from the second equation $s = \frac{2D}{k^2 + D^2}$, whence also

$$c = \frac{k^2 + D^2 - 2CD}{k^2 + D^2}, \quad d = \frac{k^2 - D^2}{k^2 + D^2}.$$

Substituting for C its value, the first-mentioned value of s becomes

$$s = \frac{(c_1 - c_2)(d_1 s_2 - d_2 s_1)}{1 - k^2 s_1^2 s_2^2 - c_1 c_2 - d_1 s_1 d_2 s_2},$$

which is a form that can be easily verified: we in fact have

$$s_1 = \operatorname{sn}(u_1 + u_2) = \frac{s_1 c_2 d_2 + s_2 c_1 d_1}{1 - k^2 s_1^2 s_2^2} = -\frac{s_1^2 - s_2^2}{s_1 c_2 d_2 - s_2 c_1 d_1} = \frac{s_1 c_1 + s_2 c_2}{c_1 c_2 + d_1 d_2} = \frac{s_1 d_1 + s_2 d_2}{d_1 d_2 + k^2 s_1 c_1 s_2 c_2},$$

see my *Elliptic Functions*, p. 63: or calling these values

$$= \frac{\mathcal{N}_1}{\mathcal{D}_1} = \frac{\mathcal{N}_2}{\mathcal{D}_2} = \frac{\mathcal{N}_3}{\mathcal{D}_3} = \frac{\mathcal{N}_4}{\mathcal{D}_4},$$

the above value is

$$s = \frac{\mathcal{N}_2 - \mathcal{N}_3}{\mathcal{D}_2 - \mathcal{D}_3},$$

which is right.

Cambridge, 28 June, 1884.

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844.

ON A THEOREM RELATING TO SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xx. (1885), pp. 212, 213.]

I FIND among my papers the following example of a general theorem relating to seminvariants.

Write

$$\begin{aligned}\Theta &= b\partial_a + c\partial_b + d\partial_c + e\partial_d + f\partial_e + g\partial_f + h\partial_g + i\partial_h, \\ \Omega &= a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + 5e\partial_f + 6f\partial_g + 7g\partial_h,\end{aligned}$$

then from a seminvariant S of the degree θ and weight w , operating upon it with $\Theta - b\delta$ and $\Omega - 2wb$, we derive two seminvariants, each of them of the degree $\theta + 1$ and weight $w + 1$; we obtain a combination of these by operating on S with $2w(\Theta - b\delta) - \delta(\Omega - 2wb)$, that is, with $2w\Theta - \delta\Omega$, and this combination is of the weight $w + 1$, but only of the degree θ ; viz. operating with $2w\Theta - \delta\Omega$, we obtain a new seminvariant of the same degree but of a weight increased by unity.

The example relates to the under-mentioned seminvariant S of the degree 3 and weight 7:

	$S =$
h	+ 1
bb	− 7
cg	+ 9
df	− 5
e^2	+12
b^2f	−30
bcc	+20
	± 42

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	ΘS	$-b\delta S$	Sum	ΩS	$-14bS$	Sum	$14\Theta S$	$-3\Omega S$	+110	Result	
i	1		1	+ 7		+ 7	+ 14	− 21	+ 7	0	
bh	− 5	− 3	− 8	− 42	− 14	− 56	− 36	+126	− 56	0	
cg	+ 2		+ 2	+ 38		+ 38	+ 28	−114	+110	+ 1	
df	+ 4		+ 4	− 2		− 2	+ 56	+ 6	−322	− 3	
e^2	− 5		− 5	− 15		− 15	− 70	+ 45	+245	+220	+ 2
b^2g	+ 3	+21	+24	+ 26	+60	+ 98	+158	+ 70	−180	−110	− 1
bcf	+ 3	−27	−24	− 96	−126	−222	+ 42	+288		+330	+ 3
bde	+ 5	+15	+20	+ 60	+70	+130	+ 70	−180		−110	− 1
c^2e	−30		−30	− 30		− 30	− 30	−420	+ 90	−330	− 3
cd^2	+20		+20	+ 20		+ 20	+280	− 60		+220	+ 2
b^3f		−36	−36				−168	−168			
b^2cf		+90	+90				+420	+420			
b^4d		−60	−60				−280	−280			
	±40	±126	±163	±185	±588	±773	±560	±553	±448	±880	±5

The columns above ΘS (where observe that, to operate with the $b\delta_a$ of Θ , we restore the proper power of a , reading S as being $= +1a^0b - 7abg + \&c.$, and putting ultimately $a = 1$) and $-b\delta S$, and the sum of these, which is a seminvariant, degree 4, weight 8; also ΩS and $-14bS$, and the sum of these, which is a seminvariant, degree 4, weight 8. They also show $14\Theta S$ and $-3\Omega S$, the sum of which would be a seminvariant, degree 3, and weight 8; instead of giving this sum, I have added a column equal to $+7(i - 8bh + 28cg - 56df + 33e^2)$, and given the sum of the three columns, which will of course be a seminvariant of the same degree and weight; the coefficients contain all of them the factor 110, and, throwing this out, we have in the last column the seminvariant $(g - 3df + \dots + 2cd^2)$ of the degree 3 and weight 8, derived by the foregoing direct process from the given seminvariant S of the degree 3 and weight 7.

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845.

ON THE ORTHOMORPHOSIS OF THE CIRCLE INTO THE PARABOLA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XX. (1885), pp. 213–220.]

It is remarked by Schwarz (see his Memoir “Ueber einige Abbildungsaufgaben,” *Crelle*, t. LXX. (1869), pp. 105–120; p. 115), that the circle $x^2 + y^2 - 1 = 0$ can be orthomorphosed into the parabola $y^2 = 4(1 - x)$ by the equation

$$\sqrt{(x' + iy')} = \tan \frac{1}{4}\pi \sqrt{(x + iy)} :$$

viz. this equation establishes a (1,1) relation between the points interior to the circle and those interior to the parabola, which, qua relation between $x + iy$ and $x' + iy'$, will be orthomorphic, that is, infinitesimal elements of the one area will correspond to similar infinitesimal elements of the other area. The diameter $y = 0$ of the circle is transformed into the axis $y' = 0$ of the parabola, and figs. 1, 2 are symmetrical on the two sides of these lines respectively; we may therefore consider only the transformation

[Figure 1: Upper semicircle ACB with centre O , diameter $A'B'$ along the horizontal axis (with A' on the left, B' on the right), arc passing through highest point C above.] [Figure 2: Upper half of parabola opening leftward; axis along the horizontal, vertex A at right, focus O , and arms extending to B above and below toward $x = -\infty$.]

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of the upper semicircle into the upper half of the parabola; we have (see figs. 1 and 2) A' corresponding to the vertex A , O' to the focus O , and B' to the point at infinity ($x = -\infty$) on the axis of the parabola; the semicircular arc $A'C'B'$ corresponds to the infinite half-arc ACB of the parabola, the highest point C' corresponding to a point C between the vertex and the semi-latus rectum.

[Figure 3: Two orthotomic systems of limaçons in the disk. One system, with B as a crunode, fills the figure as families of nested loops; the other system, with B as an acnode, gives the orthogonal trajectories.]

We may divide the circle by concentric circles and by radii; the corresponding curves for the parabola will be ovals and radials from the focus, the curves of each system being transcendental curves. Or we may divide the parabola by means of two systems of confocal parabolas; the corresponding curves for the circle will be (see fig. 3) two systems of orthotomic limaçons, those of the one system having B for a crunode, and those of the other system having B for an acnode.

To show that the circle thus corresponds to the parabola, it is only necessary to write $\sqrt{(x + iy)} = 1 + qi$, that is, $x = 1 - q^2$, $y = 2q$, implying $y^2 = 4(1 - x)$, or (x, y) is a point on the parabola; and we then obtain for $x' + iy'$ a value of the form $\cos \theta + i \sin \theta$, that is, (x', y') is a point on the circle $x'^2 + y'^2 = 1$; but in reference to what follows, I give the proof in a somewhat more artificial form.

Writing $\log \tan$ to denote the hyperbolic logarithm of the tangent, then if ϕ, ϕ' are such that

$$\phi' = \log \tan(\tfrac{1}{4}\pi + \tfrac{1}{2}\phi),$$

this equation, as is known, may also be presented in the forms

$$i\phi' = \log \tan(\tfrac{1}{4}\pi + \tfrac{1}{2}i\phi'),$$

$$i \tan \tfrac{1}{2}\phi' = \tan \tfrac{1}{2}i\phi',$$

or, what is the same thing,

$$\tan \tfrac{1}{2}\phi' = \tanh \tfrac{1}{2}\phi,$$

where observe that, as ϕ increases from 0 to $\frac{1}{2}\pi$, ϕ' increases from 0 to ∞ , and that always $\phi > \phi'$.

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We can now establish as follows the three correspondences AO with $A'O'$, OB with $O'B'$, and ACB with $A'C'B'$.

$$(1) \quad \sqrt{(x+iy)} = \frac{2\phi}{\pi}; \quad \sqrt{(x'+iy')} = \tan \frac{1}{2}\phi';$$

that is,

$$x = \frac{4\phi^2}{\pi^2}, \quad y = 0; \quad x' = \tan^2 \frac{1}{2}\phi', \quad y' = 0;$$

$$\phi = 0 \text{ gives } A, A'; \quad \phi' = \frac{1}{2}\pi \text{ gives } O, O'.$$

$$(2) \quad \sqrt{(x+iy)} = \frac{2i\phi}{\pi}; \quad \sqrt{(x'+iy')} = i \tan \frac{1}{2}\phi';$$

that is,

$$x = -\frac{4\phi^2}{\pi^2}, \quad y = 0; \quad x' = -\tan^2 \frac{1}{2}\phi', \quad y' = 0;$$

$$\phi = \phi' = 0 \text{ gives } O, O'; \quad \phi = \infty, \phi' = \frac{1}{2}\pi \text{ give } B, B'.$$

$$(3) \quad \sqrt{(x+iy)} = 1 + \frac{2i\phi}{\pi}; \quad \sqrt{(x'+iy')} = e^{i\theta};$$

that is,

$$x = 1 - \frac{4\phi^2}{\pi^2}, \quad y = \frac{4\phi}{\pi},$$

and therefore

$$y^2 = 4(1-x),$$

the parabola; and

$$x' = \cos 2\theta, \quad y' = \sin 2\theta,$$

and therefore

$$x'^2 + y'^2 = 1,$$

the circle;

$$\phi = \phi' = 0 \text{ gives } A, A'; \quad \phi = \infty, \phi' = \frac{1}{2}\pi \text{ give } B, B'.$$

Observe that for points in $O'A'$, $O'B'$, equidistant from O' , we have $x' = \tan^2 \frac{1}{2}\phi'$, $x' = -\tan^2 \frac{1}{2}\phi'$; and corresponding hereto we have points in OA , OB on the axis of the parabola, the values of x being $x = \frac{4\phi^2}{\pi^2}$, $x = -\frac{4\phi^2}{\pi^2}$, viz. the negative value is always the greater.

Observe further that, to the points $(x', y') = (\cos 2\theta, \sin 2\theta)$ and $(x'_1, y'_1) = (-\tan^2 \frac{1}{2}\phi', 0)$ on the circle, and on the radius OB' , correspond the points

$$(x, y) = \left(1 - \frac{4\phi^2}{\pi^2}, \frac{2\phi}{\pi}\right), \quad \text{and} \quad (x_1, y_1) = \left(-\frac{4\phi^2}{\pi^2}, 0\right),$$

on the parabola and on the axis OB respectively; the axial distance of these two points is $\left(1 - \frac{4\phi^2}{\pi^2}\right) - \left(-\frac{4\phi^2}{\pi^2}\right) = 1$, the radius of the circle, or the distance of the vertex and focus of the parabola; this is a rather curious theorem.

[Paper 845 continues on the following pages, which lie outside the present chunk.]